ASSIGNMENT COVER SHEET

PROGRAMME	:	Master of Business Analytics		
SUBJECT CODE AND TITLE	:	BAA5033 – Python for Business Analytics		
ASSIGNMENT TITLE	:	Individual Assignment – E-Commerce Sales Analysis Using Python		
LECTURER	:	Dr. Mohammed Muayad Abdulrazzaq Abdulrazzaq	ASSIGNMENT DUE DATE:	14/08/2025

STUDENT'S DECLARATION

1. I hereby declare that this assignment is based on my own work except where acknowledgement of sources is made.
2. I also declare that this work has not been previously submitted or concurrently submitted for any other courses in Sunway University/College or other institutions.

[Submit "Turn-it-in" report (please tick √): Yes ____ No ____]

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ADDENDUM

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Declaration

[ ] I / We used the following A.I. tools to produce content in this submission:

Tool	**Purpose**	**Prompts**	**Sections where AI output was used / Outcome(s) in the submission**
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e.g. Grammarly	*e.g. Correcting grammar and spelling, improving sentence structure*	*N/A*	*e.g. Grammarly suggestions were used for all sections of the essay*

Note: Add additional rows if necessary.

OR

[ ] I / We did not use any A.I. tools to produce any of the content in this submission.

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1.0 Introduction

This report examines an e-commerce dataset using Pandas, Matplotlib, and Seaborn to uncover insights on the products customers purchase, their shopping habits, the region driving the most sales, and consumer behaviour.

It involves five important steps as shown below:
Step 1: Examine the dataset's structure to understand its format and contents.
Step 2: Clean and explore the data to reduce errors and increase accuracy.
Step 3: Visualise key trends to identify patterns and relationships in the data.
Step 4: Improve visualisations for better interpretation.
Step 5: Analyse sales trends.

Each step throughout the assignment includes a the use of clear code, generating results, as well as explanations to make the analysis easy to follow and repeat. The main idea is to provide useful insights for making important business decisions.

2.0 Task 1: Data Overview

Here, the data set's structure was examined to understand its format and contents. The steps taken are shown in the Figures below, commencing with the python code (Figure 1) to generating the first five rows of the dataset from head() as shown in Figure 3.

Figure 1: Python code for loading the dataset

```
# Load the dataset

import pandas as pd
df = pd.read_csv("ecommerce_data.csv")
```

Figure 2: Dataset structure output from info()

```
# Display the dataset summary

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   OrderID          1000 non-null   object 
 1   ProductID        1000 non-null   object 
 2   Category         1000 non-null   object 
 3   Price            980 non-null    float64
 4   Quantity         980 non-null    float64
 5   Country          1000 non-null   object 
 6   PaymentMethod    1000 non-null   object 
 7   CustomerSegment  1000 non-null   object 
 8   OrderDate        1000 non-null   object 
 9   TotalSales       1000 non-null   float64
dtypes: float64(3), object(7)
memory usage: 78.3+ KB
```

Figure 3: First five rows of the dataset from head()

```
# Display the first five rows of the dataset
df.head()
```

	OrderID	ProductID	Category	Price	Quantity	Country	PaymentMethod	CustomerSegment	OrderDate	TotalSales
0	O00001	P0103	Electronics	25.58	5.0	UK	Bank Transfer	Wholesale	2018-11-21	127.90
1	O00002	P0271	Beauty	78.43	5.0	Canada	PayPal	Wholesale	2017-05-15	392.15
2	O00003	P0107	Beauty	493.38	5.0	Canada	Cash on Delivery	Regular	2020-11-23	2466.90
3	O00004	P0072	Sports	482.73	5.0	India	Cash on Delivery	Wholesale	2018-01-13	2413.65
4	O00005	P0189	Home & Kitchen	7.45	3.0	Australia	Bank Transfer	Regular	2015-08-02	22.35

The dataset consisted of 1,000 rows of transactions and 10 columns, including OrderID, ProductID, Category, Price, Quantity, Country, PaymentMethod, CustomerSegment, OrderDate, and TotalSales as shown in Figure 3 above. Initial checks showed the data types and revealed 20 missing values in Price and Quantity columns using info() as shown in row 3 and 4 of Figure 2. The head() output provided a clear view of the data structure. These results will guide our cleaning process as progress is made through the assignment.

3.0 Task 2: Data Analysis

In this step, the data is analysed with a view to explore and clean the data to reduce errors and increase accuracy and relevance of the data. The Figures below show the steps taken in the process. It commences with the counting of the unique product categories and dataset dimensions (Figures 4 and 5) to displaying the total number of transactions (Figure 6), with identifying the missing values per columns (Figure 7), counting and showing the transactions with negative total sales (Figures 8 and 9), cleaning the data and performing a validation check (Figures 10 and 11), and product quantity sales and transactions (Figures 12 and 13).

Figure 4: Count of unique product categories

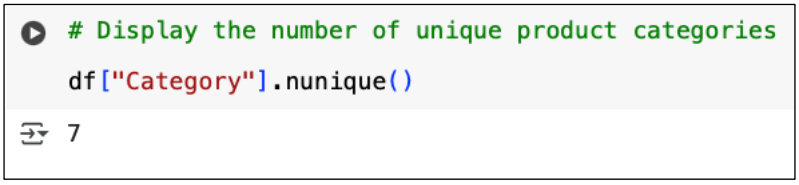


Figure 5: Dataset dimensions

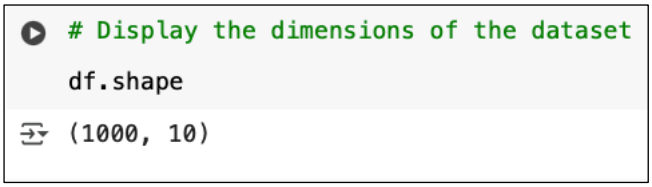


Figure 6: Total number of transactions

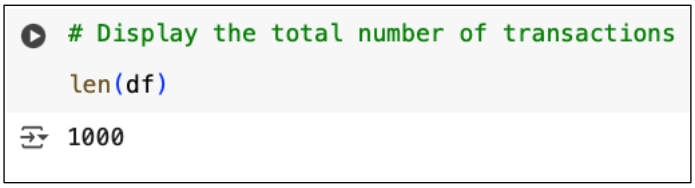


Figure 7: Missing values per column

```
# Display missing values per column
df.isnull().sum()
```

	0
OrderID	0
ProductID	0
Category	0
Price	20
Quantity	20
Country	0
PaymentMethod	0
CustomerSegment	0
OrderDate	0
TotalSales	0

dtype: int64

Figure 8: Count of transactions with negative TotalSales

```
# Display the number of transactions with negative TotalSales
(df["TotalSales"] < 0).sum()
```

```
np.int64(10)
```

Figure 9: Transactions with negative TotalSales

```
# Display rows with negative TotalSales
df[df["TotalSales"] < 0]
```

	OrderID	ProductID	Category	Price	Quantity	Country	PaymentMethod	CustomerSegment	OrderDate	TotalSales
139	O00140	P0128	Home & Kitchen	122.25	2.0	India	PayPal	Wholesale	2023-02-10	-100.0
307	O00308	P0176	Toys	309.53	8.0	Canada	PayPal	Wholesale	2017-02-10	-100.0
349	O00350	P0147	Home & Kitchen	393.58	1.0	Canada	Bank Transfer	Wholesale	2021-11-26	-100.0
371	O00372	P0272	Electronics	341.71	9.0	Australia	Cash on Delivery	Wholesale	2022-07-18	-100.0
455	O00456	P0268	Beauty	256.64	8.0	UK	PayPal	Wholesale	2015-01-25	-100.0
526	O00527	P0019	Books	167.46	7.0	Japan	Bank Transfer	Regular	2020-06-15	-100.0
587	O00588	P0253	Electronics	246.35	8.0	Japan	Cash on Delivery	Premium	2017-11-27	-100.0
910	O00911	P0261	Beauty	303.88	1.0	Canada	Bank Transfer	Regular	2015-09-28	-100.0
913	O00914	P0173	Sports	195.00	9.0	Japan	PayPal	Premium	2018-07-04	-100.0
999	O01000	P0285	Home & Kitchen	16.56	3.0	Australia	Cash on Delivery	Premium	2022-11-29	-100.0

Figure 10: Removing negative TotalSales records

```
# Remove rows with negative TotalSales
df_clean = df[df["TotalSales"] >= 0].copy()
```

Figure 11: Validation check after cleaning

```
# Check for negative TotalSales after cleaning
(df_clean["TotalSales"] < 0).sum()
```

```
np.int64(0)
```

Figure 12: Top 3 products by quantity sold

```
# Display the top 3 most sold products by quantity
df_clean.groupby("ProductID")['Quantity'].sum().sort_values(ascending=False).head(3)
```

ProductID	Quantity
P0220	46.0
P0099	46.0
P0252	43.0

dtype: float64

Figure 13: Product category transaction count

```
# Display the product category with the most transactions
df_clean["Category"].value_counts()
```

Category	count
Clothing	151
Beauty	148
Electronics	142
Home & Kitchen	140
Sports	137
Books	137
Toys	135

dtype: int64

The data used in the study consisted of 1,000 transactions (Figure 5). In the analysis, the data was arranged into 10 columns (Figure 5). The columns were composed of 3 numerical columns and 7 columns were categorical in nature. When the initial data analysis was performed, 7 distinct product categories were recognised. The analysis also revealed 20 missing values which were discovered in each of the Price and Quantity columns. The TotalSales value showed 10 transactions that had negative values (Figure 9). These values were then ignored and removed to ensure that the data was accurate, reliable and relevant.

After the cleaning process, no negative sales values remained in the dataset (Figure 11). The analysis revealed that the Clothing category had the highest number of transactions, 151, while the top three best-selling products based on quantity were P0220, P0099, and P0252 (Figures 12 and 13).

4.0 Task 3: Visualisation

In this section, the idea was to get a clearer visual of the total units sold based on the product categories. For this the data visualisation tool was loaded (Figure 14).

Figure 14: Importing Matplotlib for data visualisation.

```
# Load data visualisation tools
import matplotlib.pyplot as plt
```

The subsequent steps to get clearer visuals are shown below and where some explanations with regard to the visual is provided accordingly (Figure 15 to Figure 17).

Figure 15 below shows the horizontal bar chart which details the product categories based on the sales volume. It can be seen that the product category generating the highest sales volume was Beauty, followed by Clothing and Electronics. The sales volume of Toys, Sports, and Home and Kitchen, did not show much difference. The lowest sales volume was attributed to Books.

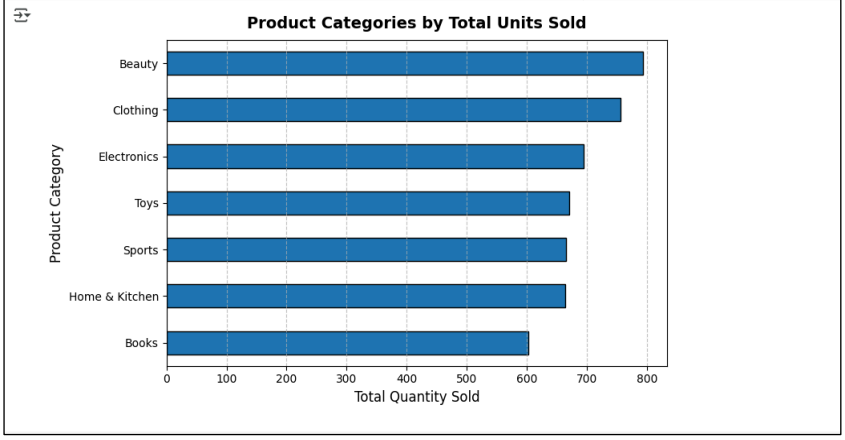
Figure 15: Product categories by total units sold

```
# Horizontal bar chart: Product category with the most sales volume

(df.groupby('Category')['Quantity'].sum()
 .sort_values()
 .plot(kind='barh',
       color='#1f77b4',
       edgecolor='black',
       figsize=(8,5)))

plt.title('Product Categories by Total Units Sold', fontsize=14, fontweight='bold', pad=10)
plt.xlabel('Total Quantity Sold', fontsize=12)
plt.ylabel('Product Category', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()

plt.show()
```



Subsequently, a line chart showing the number of orders from 2015 to 2023 was generated.

Figure 16: Number of orders by year

```
# Line chart: Year with the most number of orders

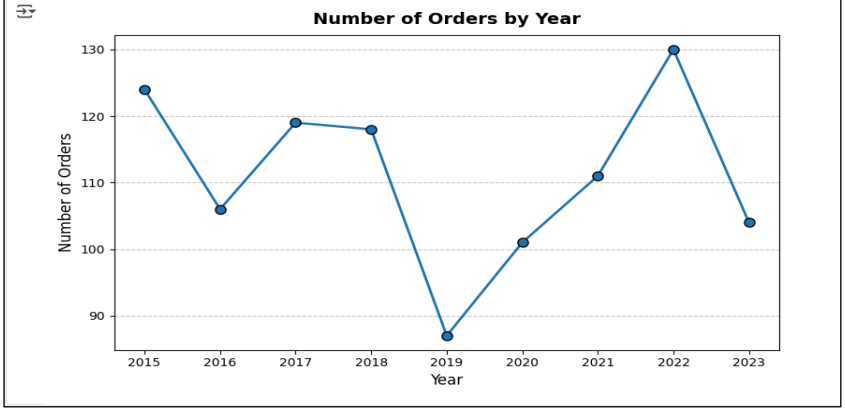
df['Year'] = pd.to_datetime(df['OrderDate']).dt.year

orders_per_year = df['Year'].value_counts().sort_index()

plt.figure(figsize=(8,5))
plt.plot(orders_per_year.index, orders_per_year.values,
         marker='o', markersize=8, markeredgecolor='black',
         markerfacecolor='#1f77b4', linewidth=2)

plt.title('Number of Orders by Year', fontsize=14, fontweight='bold', pad=10)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Number of Orders', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.7)

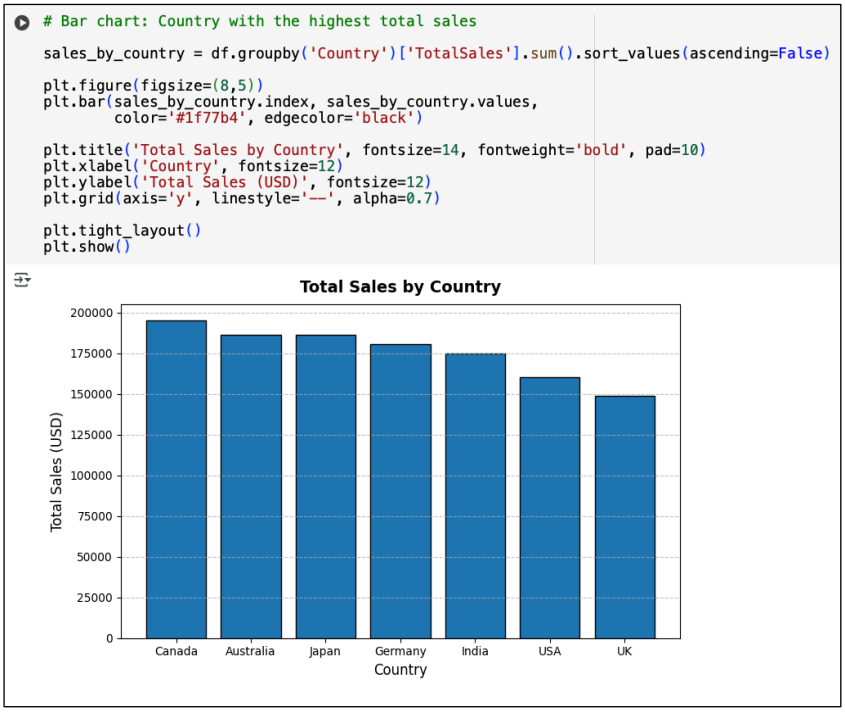
plt.tight_layout()
plt.show()
```



From Figure 16 above, it can be seen that the order volumes appear to vary from year to year. It is observed that a significant drop in order volume occurred in 2019 (the lowest number of orders) followed by a steady increase and reaching a peak in 2022 (the highest number of orders). However, a subsequent drop in orders was then seen in 2023.

The total sales based on a country basis was generated (see Figure 17 below). The largest sales were seen in Canada (approximately USD 195,337). There was little difference between the countries in the 2nd to 4th position, namely, Australia, Japan and Germany. The lowest total sales were recorded in the UK.

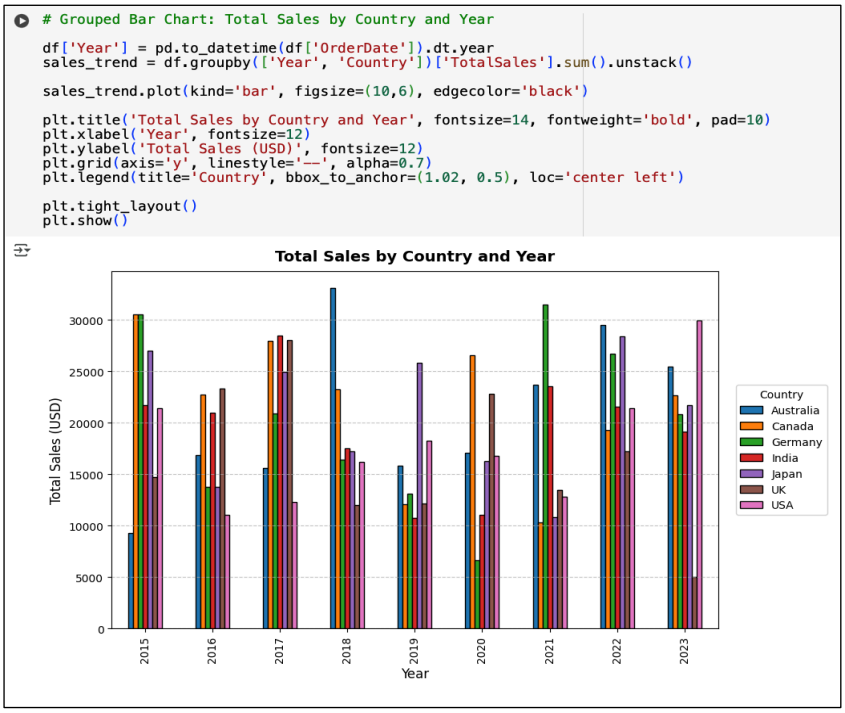
Figure 17: Total sales by country



5.0 Task 4: Plot Evaluation and Improvement

The bar chart from Figure 17 in Task 3 showed total sales across countries but it is difficult to see the overall yearly totals and the trend over the years on a comparative basis . It was decided that a grouped bar chart of the countries over the years will provide a more detailed picture (Figure 18).

Figure 18: Grouped Bar Chart – Total Sales by Country and Year



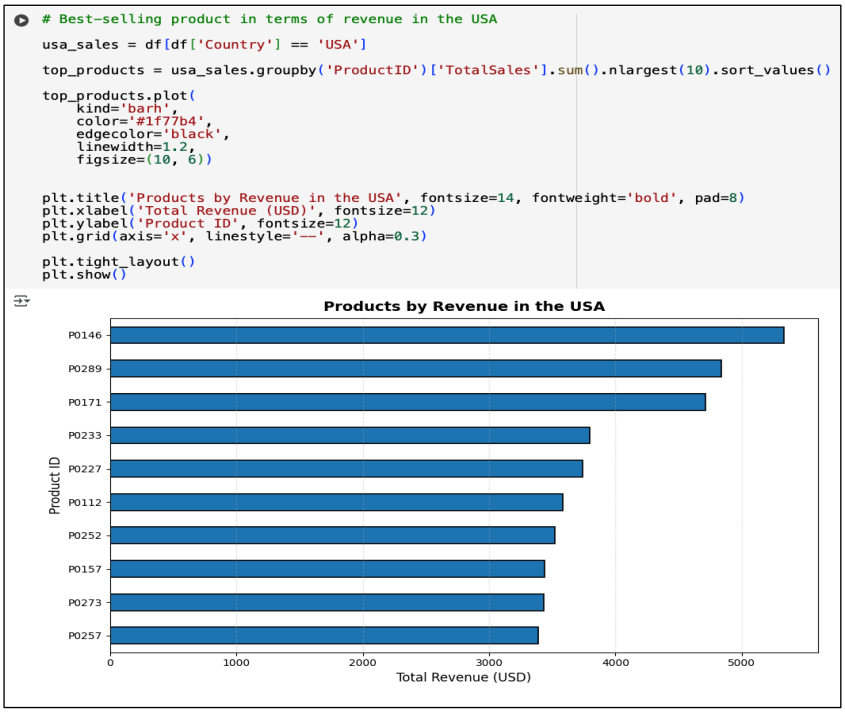
The improved stacked bar chart in Figure 18 shows both the yearly totals and the proportional contributions of each country on a yearly basis. Key improvements in the plot include the stacked format for clearer year-by-year composition, a refined colour palette to improve country distinction, and repositioning the legend outside the plot to reduce clutter.

The stacked pattern appears to vary from year to year. For example, in 2015, the top three countries for sales were Canada, Japan and Germany. In 2016 and 2017 the top three sales contributors were Canada, India and the UK. In 2022, the top three were the USA, Australia and Canada. Other observations show that there are one-off peaks experiences, like Australia (2018), Japan (2019), Germany (2021), the USA (in 2023). The grouped bar chart makes it easier to understand the contribution by the different countries to the sales for any one year, and also provides a view to the variations from year to year when compared to the original bar chart in Figure 17.

6.0. Task 5: Sales Trend Analysis

In order to get a clearer picture of the products contributing to the sales, an analysis of the products sold in the USA was selected. The is shown in the bar chart in Figure 19 below.

Figure 19: Best-Selling Product in the USA




```
# Best-selling product in terms of revenue in the USA

usa_sales = df[df['Country'] == 'USA']

top_products = usa_sales.groupby('ProductID')['TotalSales'].sum().nlargest(10).sort_values()

top_products.plot(
    kind='barh',
    color='#1f77b4',
    edgecolor='black',
    linewidth=1.2,
    figsize=(10, 6))


plt.title('Products by Revenue in the USA', fontsize=14, fontweight='bold', pad=8)
plt.xlabel('Total Revenue (USD)', fontsize=12)
plt.ylabel('Product ID', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)

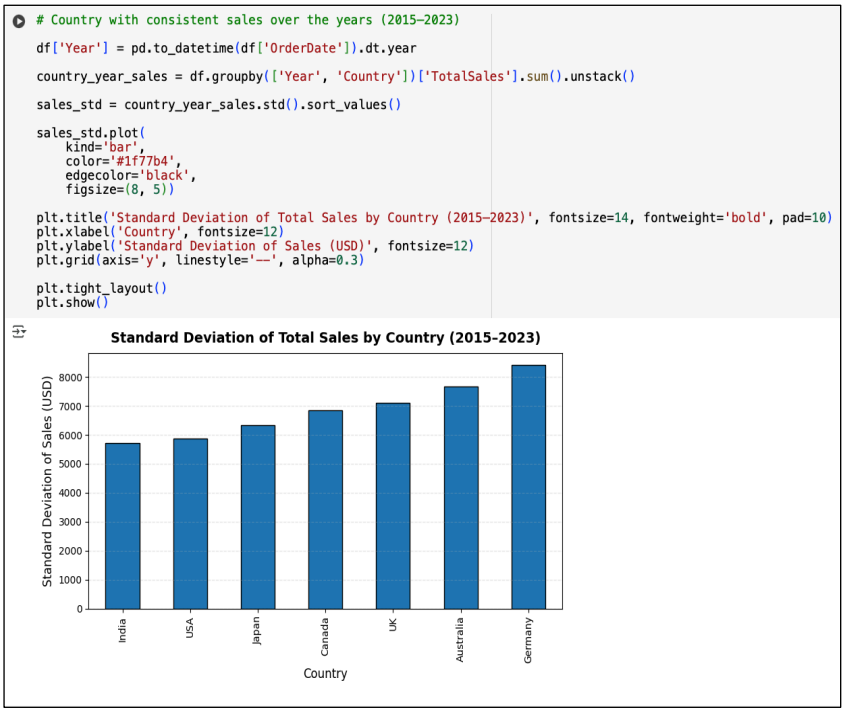
plt.tight_layout()
plt.show()
```

The chart exhibits the top 10 products by revenue in the USA. The highest revenue earner was Product P0146, making it the best-selling product in the USA. It can be seen that the other strong performers were Products P0289 and P0171. From the chart products, P233 and P227 are generating almost similar revenue. As for the remaining products, it is observed that they do not vary significantly in their total revenues but cumulatively they do contribute significantly to the total revenue.

Another indicator of sales trends was to look at the standard deviation of total yearly sales which gives an indication of the consistency of sales over the years. Figure 20 below shows the standard deviation of total sales by country from 2015 to 2023.

Germany had the highest standard deviation, indicating greater fluctuation and volatility in annual sales throughout the 2015–2023 period, compared to India which showed the greatest stability based on the lowest value for its standard deviation. For the remaining countries from Australia, UK, Canada, Japan and the USA, the standard deviation begins to fall showing a reducing market volatility from Australia to the USA.

Figure 20: Countries with Consistent Sales (2015–2023)




```
# Country with consistent sales over the years (2015–2023)

df['Year'] = pd.to_datetime(df['OrderDate']).dt.year

country_year_sales = df.groupby(['Year', 'Country'])['TotalSales'].sum().unstack()

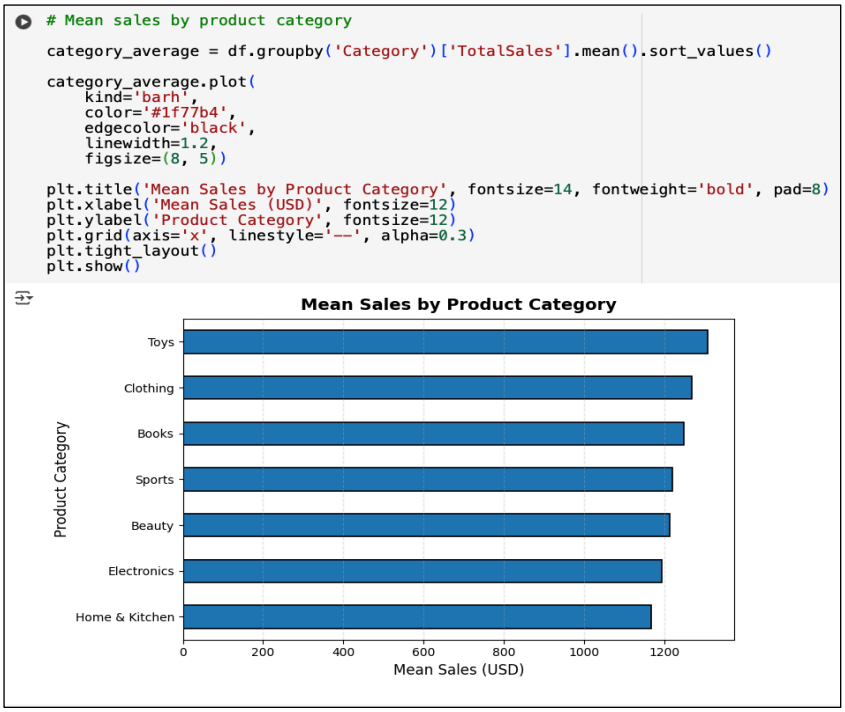
sales_std = country_year_sales.std().sort_values()

sales_std.plot(
    kind='bar',
    color='#1f77b4',
    edgecolor='black',
    figsize=(8, 5))

plt.title('Standard Deviation of Total Sales by Country (2015–2023)', fontsize=14, fontweight='bold', pad=10)
plt.xlabel('Country', fontsize=12)
plt.ylabel('Standard Deviation of Sales (USD)', fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.3)

plt.tight_layout()
plt.show()
```

Figure 21: Mean Sales by Product Category




```
# Mean sales by product category

category_average = df.groupby('Category')['TotalSales'].mean().sort_values()

category_average.plot(
    kind='barh',
    color='#1f77b4',
    edgecolor='black',
    linewidth=1.2,
    figsize=(8, 5))

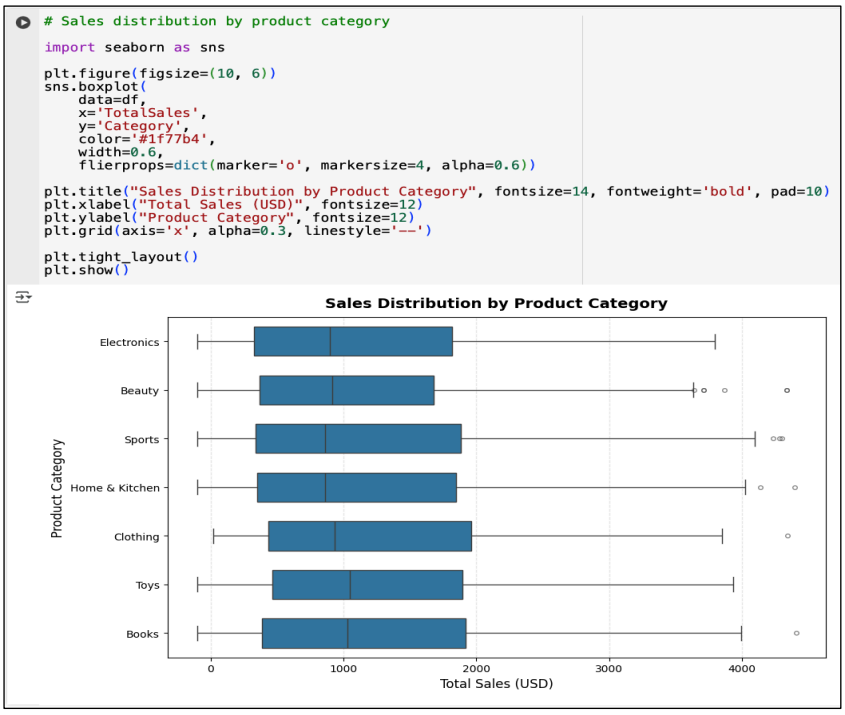
plt.title('Mean Sales by Product Category', fontsize=14, fontweight='bold', pad=8)
plt.xlabel('Mean Sales (USD)', fontsize=12)
plt.ylabel('Product Category', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.3)
plt.tight_layout()
plt.show()
```

The above chart shows that there is no major or significant difference in the mean sales per category. This implies that there is no one outstanding or leading product. While Toys and Clothing slightly outperform the other categories, on average the difference is not

significant. This shows that no single product category dominates overall sales, and all categories play an important role in generating sales and revenue.

Finally, the boxplot was undertaken to determine the spread and variability in sales within each product category. It is seen that the median sales of the product categories are not significantly different, thus, showing that there is no one exceptional product contributing to the overall sales. The boxes are also about the same length but the variability is shown by the whiskers and outliers. However, the product with the furthest right outlier which is indicative of the extreme variability in orders is Sports, which is then followed by Home and Kitchen. These spikes are contributed by bulk purchases, sales due to promotions or premium items within the respective categories. Electronics and Toys, which have no outliers, are the most stable products and their sales are also generally predictable.

Figure 22: Sales Distribution by Product Category




```
# Sales distribution by product category

import seaborn as sns

plt.figure(figsize=(10, 6))
sns.boxplot(
    data=df,
    x='TotalSales',
    y='Category',
    color='#1f77b4',
    width=0.6,
    flierprops=dict(marker='o', markersize=4, alpha=0.6))

plt.title("Sales Distribution by Product Category", fontsize=14, fontweight='bold', pad=10)
plt.xlabel("Total Sales (USD)", fontsize=12)
plt.ylabel("Product Category", fontsize=12)
plt.grid(axis='x', alpha=0.3, linestyle='--')

plt.tight_layout()
plt.show()
```

7.0 Conclusion

This e-commerce analysis highlights that Beauty products led sales, whereas Books underperformed, indicating a potential need for inventory adjustment. Canada emerged as the top revenue generator, whereas the UK has potential to improve. India showed stable

performance, while Germany showed the greatest variability in total sales. The grouped bar chart on total sales from 2015 to 2023 showed that while the sales were being generated in the current markets, but the markets were not entirely consistent with different leaders appearing at different times, such as, Canada in 2015 and 2020, UK in 2016, India in 2017, Australia in 2018, Japan in 2019, Germany in 2021 and the US in 2023. There was no clear market leader. Opportunities lie with the company to further study the forces driving sales of products and try to create a more level revenue generation from the different market to enhance total revenue.

The U.S. showed strong growth opportunities, particularly through its best-selling product (P0146). Similar product by revenue analysis should be done for other countries to identify the best-selling products and to find opportunities for promoting and increasing the underperforming products. While the mean sales by product category does not vary significantly and this was corroborated by the sales distribution by product category, the boxplots showed that there was extreme variability in orders in the Sports and Home and Kitchen categories with their outliers. These spikes, as indicated earlier, are contributed by bulk purchases, sales due to promotions or premium items within the respective categories. Electronics and Toys, which have no outliers, were the most stable products and their sales are also generally predictable.

Despite a sales dip in 2019, 2022 marked a significant peak, underscoring the importance of resilience in the market. The dip in 2023 requires further analysis to create a turnaround just like the period from 2019 to 2022.

In conclusion, by leveraging on Python analytics and generating clear visualisations, the company has the ability and capacity to create insights that provide valuable guidance for identifying high-performing products, targeting profitable markets, and, optimising and stabilizing operations for long-term, sustainable growth.

8.0 References

Code and analysis are available in the Google Colab Notebook.